Course Name: Geographic Information System

Week 05 - Assignment 05 - (Jul-2026)

TYPE OF QUESTIONS: MCQ/MSQ

Number of questions: 10 **Total marks: 10x1= 10**

QUESTION 1

_____ refers to how closely repeated measurements agree with each other.

a) Accuracy
b) Precision
c) Completeness
d) Consistency

Answer: (b)

Detailed Solution:

Precision measures the consistency of repeated measurements, regardless of whether they are close to the true value.

QUESTION 2

Which factor can reduce the reliability of GIS spatial data?

a) Outdated datasets
b) Poor spatial resolution
c) Incorrect collection methods
d) All of the above

Answer: (d)

Detailed Solution:

Reliability decreases when data is old, collected improperly, or has low spatial resolution.

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QUESTION 3

An error matrix in remote sensing is mainly used for evaluating _____.

a) Coordinate systems
b) Image classification results
c) Raster compression
d) Projection accuracy

Answer: (b)

Detailed Solution:

Error matrices compare classified images with reference data to assess classification accuracy.

QUESTION 4

Producer's accuracy in a confusion matrix is calculated using _____.

a) Row totals
b) Column totals
c) Total pixels
d) Non-diagonal elements

Answer: (b)

Detailed Solution:

Producer's accuracy is the ratio of correctly classified pixels to the column total.

QUESTION 5

Incorrect inclusion of pixels into a category is known as _____.

a) Omission error
b) Commission error
c) Projection error
d) Topology error

Answer: (b)

Detailed Solution:

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Commission error occurs when pixels are wrongly assigned to a class where they do not belong.

QUESTION 6

Which of the following represents a topology error in vector GIS?

a) Properly connected nodes
b) Closed polygons
c) Overshoots
d) Georeferenced raster

Answer: (c)

Detailed Solution:

Overshoots occur when vector lines extend beyond their intended intersection points.

QUESTION 7

Reducing the number of unnecessary vertices in a line feature is called _____.

a) Line densification
b) Line simplification
c) Line buffering
d) Polygon clipping

Answer: (b)

Detailed Solution:

Line simplification reduces coordinate points while preserving the general shape of the feature.

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QUESTION 8

The alignment of two raster datasets into the same coordinate framework is called _____.

a) Buffering
b) Digitization
c) Coregistration
d) Reclassification

Answer: (c)

Detailed Solution:

Coregistration ensures multiple raster datasets spatially align correctly.

QUESTION 9

Which transformation method stretches a map layer to fit predefined control points?

a) Geocoding
b) Clipping
c) Rubber sheeting
d) Buffering

Answer: (c)

Detailed Solution:

Rubber sheeting adjusts spatial data geometrically to match reference control points.

QUESTION 10

In digital image processing, the dominant wavelength or actual color is represented by _____.

a) Brightness
b) Intensity
c) Saturation
d) Hue

Answer: (d)

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Detailed Solution:

Hue defines the actual perceived color such as red, green, or blue.

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